```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from google.colab.patches import cv2_imshow
```

```
uploaded = files.upload()
```

Choose Files satellite.jpg

satellite.jpg(image/jpeg) - 173398 bytes, last modified: 9/10/2026 - 100% done

```
Saving satellite.jpg to satellite.jpg
```

```
filename = list(uploaded.keys())[0]

print("Uploaded file:", filename)
```

```
Uploaded file: satellite.jpg
```

```
image = cv2.imread(filename)

print("Image loaded successfully!")
print("Image shape:", image.shape)
```

```
Image loaded successfully!
Image shape: (810, 1440, 3)
```

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
plt.figure(figsize=(8,6))
plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')
plt.show()
```

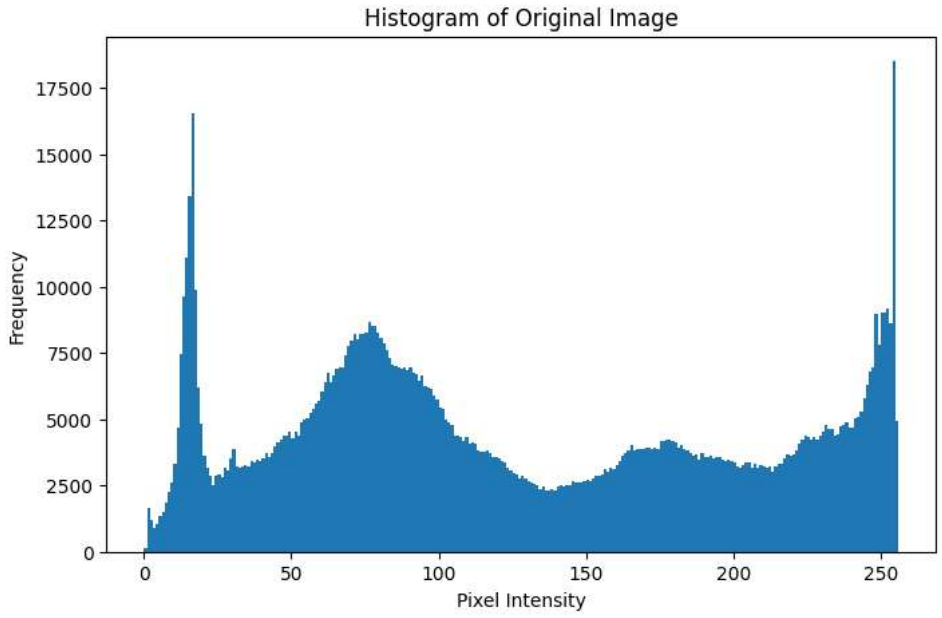



```
plt.figure(figsize=(8,5))

plt.hist(gray.ravel(), bins=256, range=[0,256])

plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.show()
```




```
min_val = np.min(gray)
max_val = np.max(gray)

contrast_stretched = ((gray - min_val) / (max_val - min_val) * 255).astype(np.uint8)
```

```
plt.figure(figsize=(8,6))

plt.imshow(contrast_stretched, cmap='gray')
plt.title("Contrast Stretched Image")
plt.axis('off')

plt.show()
```




```
print("Original intensity range:",
      np.min(gray), "to", np.max(gray))

print("After contrast stretching:",
      np.min(contrast_stretched), "to", np.max(contrast_stretched))
```

```
Original intensity range: 0 to 255
After contrast stretching: 0 to 255
```

```
print("Original intensity range:",
      np.min(gray), "to", np.max(gray))

print("After contrast stretching:",
      np.min(contrast_stretched), "to", np.max(contrast_stretched))
```

```
Original intensity range: 0 to 255
After contrast stretching: 0 to 255
```

```
print("Original intensity range:",
      np.min(gray), "to", np.max(gray))

print("After contrast stretching:",
      np.min(contrast_stretched), "to", np.max(contrast_stretched))
```

```
Original intensity range: 0 to 255
After contrast stretching: 0 to 255
```

```
hist_equalized = cv2.equalizeHist(gray)
```

```
plt.figure(figsize=(8,6))

plt.imshow(hist_equalized, cmap='gray')
plt.title("Histogram Equalized Image")
plt.axis('off')

plt.show()
```

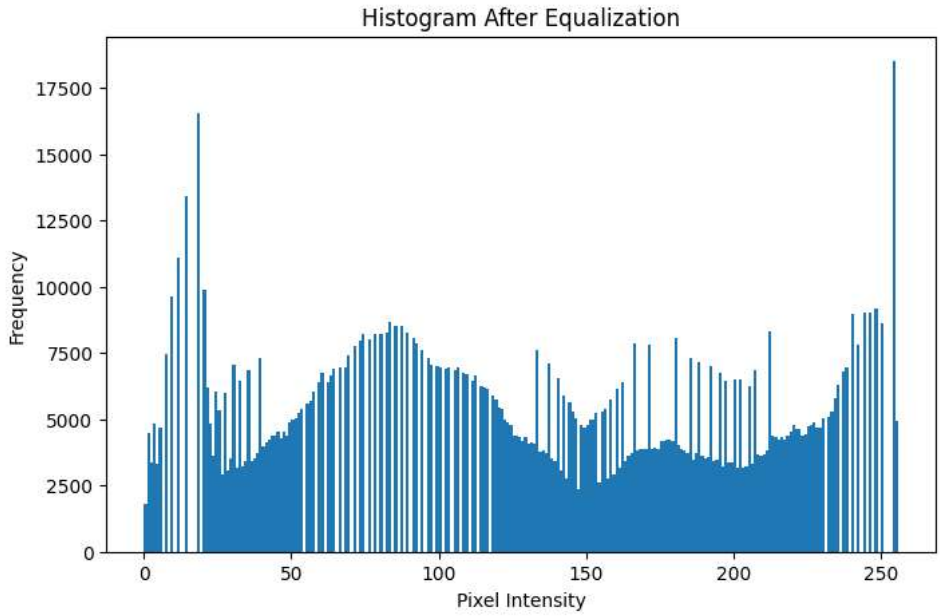



```
plt.figure(figsize=(8,5))

plt.hist(hist_equalized.ravel(),
         bins=256,
         range=[0,256])

plt.title("Histogram After Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.show()
```




```
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8,8)
)
```

```
clahe_image = clahe.apply(gray)
```

```
plt.figure(figsize=(8,6))

plt.imshow(clahe_image, cmap='gray')
plt.title("CLAHE Enhanced Image")
plt.axis('off')

plt.show()
```




```
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(hist_equalized, cmap='gray')
plt.title("Histogram Equalization")
plt.axis('off')
```

```
plt.subplot(1,3,3)
plt.imshow(clahe_image, cmap='gray')
plt.title("CLAHE")
plt.axis('off')

plt.tight_layout()
plt.show()
```



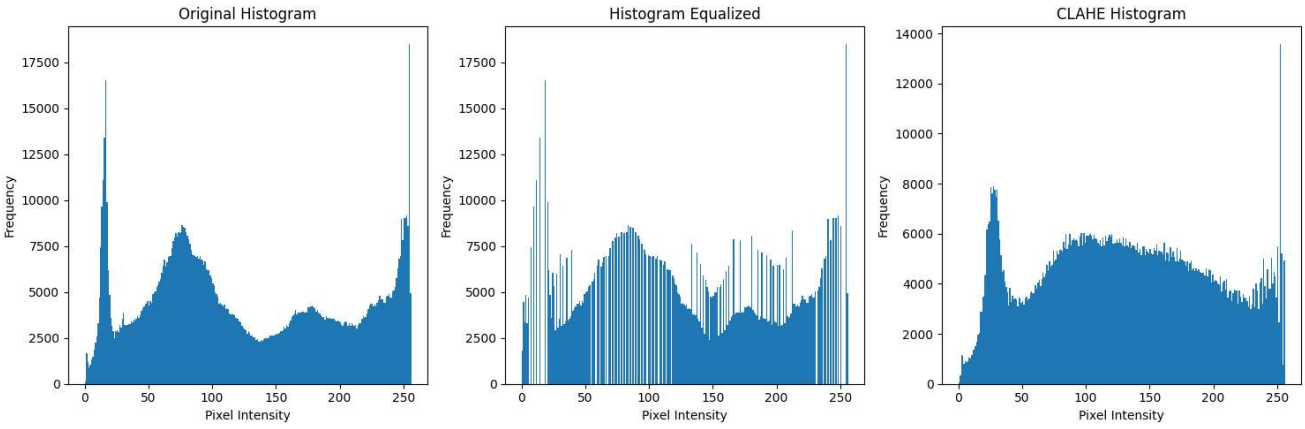

```
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.hist(gray.ravel(), bins=256, range=[0,256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,2)
plt.hist(hist_equalized.ravel(), bins=256, range=[0,256])
plt.title("Histogram Equalized")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1,3,3)
plt.hist(clahe_image.ravel(), bins=256, range=[0,256])
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(15,10))

# Original image
plt.subplot(2,3,1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis('off')

# Original histogram
plt.subplot(2,3,4)
plt.hist(gray.ravel(), bins=256, range=[0,256])
plt.title("Original Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

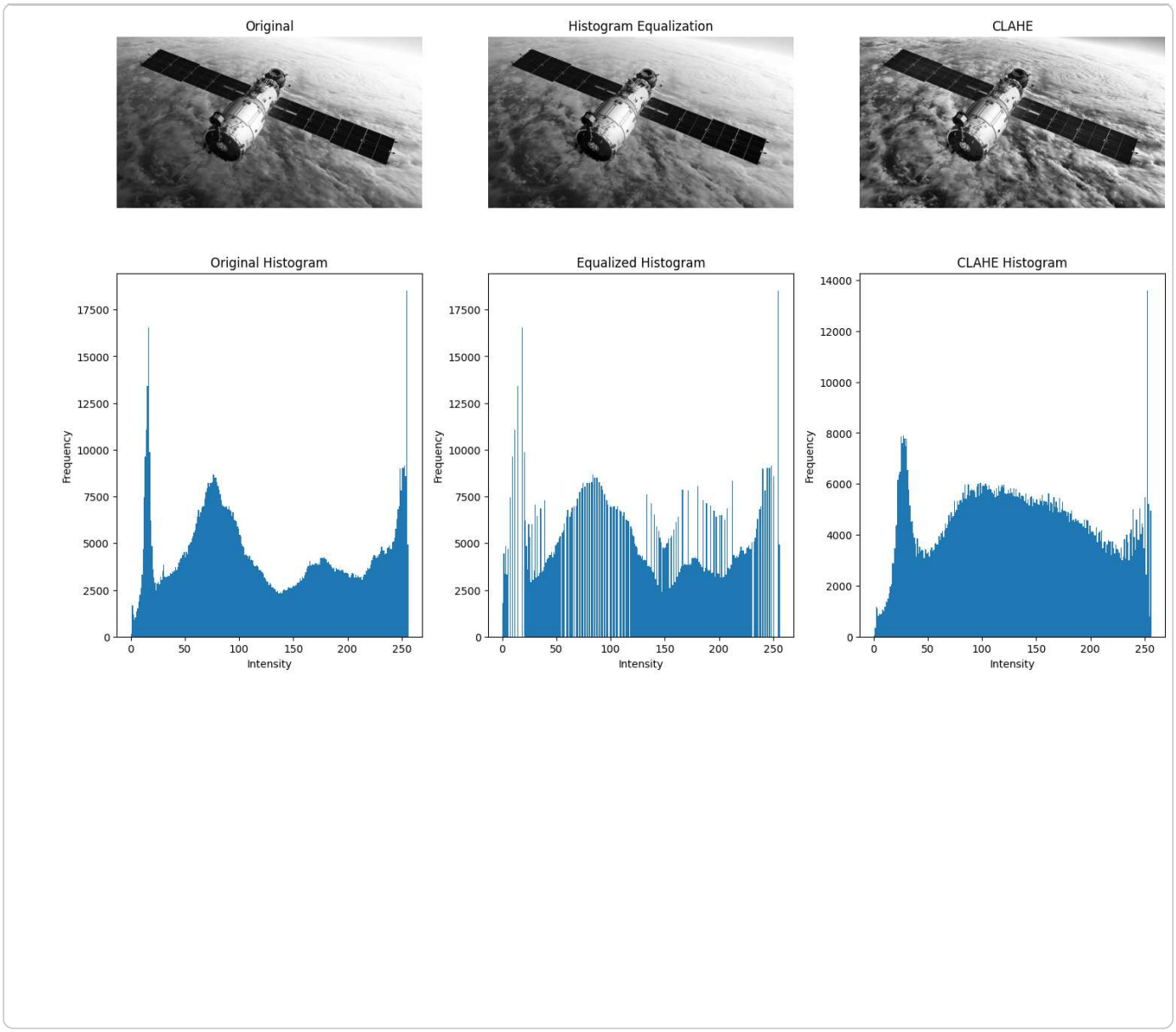
# Histogram equalized
plt.subplot(2,3,2)
plt.imshow(hist_equalized, cmap='gray')
plt.title("Histogram Equalization")
plt.axis('off')

# Equalized histogram
plt.subplot(2,3,5)
plt.hist(hist_equalized.ravel(), bins=256, range=[0,256])
plt.title("Equalized Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

# CLAHE
plt.subplot(2,3,3)
plt.imshow(clahe_image, cmap='gray')
plt.title("CLAHE")
plt.axis('off')

# CLAHE histogram
plt.subplot(2,3,6)
plt.hist(clahe_image.ravel(), bins=256, range=[0,256])
plt.title("CLAHE Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



PART B — THEORY QUESTIONS Q1. What is image contrast, and why is it important in image processing?

Answer:

Image contrast refers to the difference in intensity between different regions of an image. High contrast makes objects and details easier to distinguish, while low contrast can make an image appear dull and difficult to analyze.

Contrast is important because it improves visibility and helps computer vision systems identify important features.

Q2. Explain the concept of an image histogram. What information does it provide?

Answer:

An image histogram is a graphical representation of the distribution of pixel intensity values in an image.

It shows:

Pixel intensity values on the X-axis. Number of pixels at each intensity on the Y-axis.

It helps analyze the brightness, contrast and intensity distribution of an image.

Q3. Differentiate between Histogram Stretching and Histogram Equalization. Histogram Stretching Histogram Equalization

Expands the existing intensity range Redistributes intensity values Maps minimum and maximum values to a wider range Uses the cumulative distribution of intensities Relatively simple technique More advanced enhancement technique Preserves the general distribution shape Can significantly alter the distribution Useful for images with a limited dynamic range Useful for improving overall contrast

Q4. What is CLAHE? How does it differ from standard Histogram Equalization?

Answer:

CLAHE stands for Contrast Limited Adaptive Histogram Equalization.

Unlike standard Histogram Equalization, which operates globally on the entire image, CLAHE divides the image into small regions called tiles and performs histogram equalization locally.

CLAHE also limits contrast enhancement using a clip limit, which helps prevent excessive amplification of noise.

Therefore, CLAHE is particularly useful when different areas of an image have different illumination conditions.

Q5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Answer:

Histogram equalization improves image contrast and makes important structures and boundaries more visible.

This can help:

Improve feature visibility. Make edges easier to detect. Improve segmentation. Increase the distinction between different regions. Provide better input for subsequent image analysis.

Q6. Mention three real-world applications where histogram equalization is widely used.

Answer:

Three applications are:

Medical imaging — improving visibility of structures in medical images. Satellite imagery — enhancing details and terrain information. Low-